# Writing & Interpreting Linear Models

## WARM UPS

TEACHER COPY | Display student pages; keep solutions private

### 1 | Solve

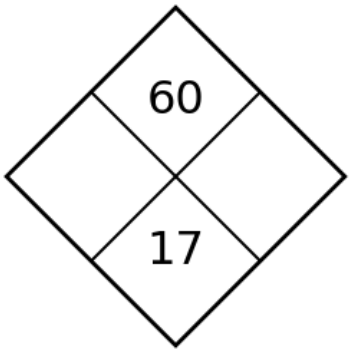
Solve for x. Show each step, then check your answer.

$$2(3x - 4) + 5 = 27$$

> **ANSWER / TEACHING NOTE**
> 2(3x - 4) + 5 = 27 → 6x - 8 + 5 = 27 → 6x = 30 → x = 5.

### 2 | Diamond

Complete the diamond.



# Writing & Interpreting Linear Models

## WARM UPS

### 3 | Spiral Review

Calculate slope between (1,7) and (3,15).

> **ANSWER / TEACHING NOTE**
> m = (15 - 7)/(3 - 1) = 8/2 = 4.

### 4 | Spiral Review

Is {(0,2),(1,4),(2,6)} a function? List domain and range.

> **ANSWER / TEACHING NOTE**
> Yes. Domain {0,1,2}; range {2,4,6}.

# Writing & Interpreting Linear Models

## LEARNING FOCUS & NOTES

**Learning focus | LG 03-ican-2 / LG 03-ican-5**

- I can write y = mx + b from a description, table, or two points.
- I can explain what the slope and starting value mean.
- I can check whether a model makes sense for a stated input.

## Vocabulary | Highlight the important words

### Linear model

An equation, table, or graph using constant change to represent a situation.

### Slope-intercept form

The equation y = mx + b, where m is slope and b is the output at x = 0.

### Slope / rate m

The change in output for each 1-unit input increase; include units.

### y-intercept / starting value b

The value of y when x = 0; often the starting amount in context.

### Variable

A letter representing a changing or unknown quantity, with an assigned meaning.

### Domain restriction

A limit on which inputs make sense, based on the situation or mathematical rule.

## Meaning & Method | Rewritten review notes

**1.** Read the story first: choose an input x and an output y with meaningful units.

**2.** Identify the starting amount b at input zero, then identify the constant change m per one input unit. Write y = mx + b.

**3.** Check the model using an input and output from the context. State restrictions such as nonnegative whole-number counts.

These are original, concise review notes aligned to the course lessons; they are not a reproduction of the textbook pages.

# Writing & Interpreting Linear Models

## EXAMPLES & YOU TRY ITS

### Example

A printing service charges a $12 setup fee plus $3 per poster. Write the cost equation, explain the fee and per-poster rate, and find the cost for 5 posters.

| Posters p | Cost C ($) |
|---|---|
| 0 | 12 |
| 1 | 15 |
| 2 | 18 |

**Write the cost equation. Explain the fees and find the cost of 5 posters.**

> **ANSWER / TEACHING NOTE**
> b = $12 setup fee, m = $3/poster. C = 3p + 12; for p=5, C = $27. Count p is a nonnegative integer.

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### You Try It

A studio charges a $9 reservation fee plus $4 per hour. Complete the table, write the cost equation, explain the slope and starting value, and calculate the cost for 3 hours.

| Hours h | Cost C ($) |
|---|---|
| 0 | ___ |
| 1 | ___ |
| 2 | ___ |

**Write the equation, interpret m and b, find cost for 3 hours.**

> **Write the equation, interpret the $9 and $4, and find cost for 3 hours.**
> C = 4h + 9; b = $9 reservation fee; m = $4/hour; cost for 3 hours is $21.

**How do the setup fee and rate appear in your equation?**

> **TEACHING NOTE**
> The starting fee is the value at zero (intercept); the hourly or per-item charge is the slope.

# Writing & Interpreting Linear Models

## LAUNCH

### Which rental plan fits?

Two bike rental plans are offered. Plan A costs $6 plus $4 per hour. Plan B costs $10 plus $3 per hour. Compare their costs at the times shown in the table.

| Hours h | Plan A ($) | Plan B ($) |
|---|---|---|
| 0 | ___ | ___ |
| 1 | ___ | ___ |
| 2 | ___ | ___ |
| 3 | ___ | ___ |

**Plan A:** $A = 4h + 6$

**Plan B:** $B = 3h + 10$

Use the table to make your comparison for these specific times.

**1 | What does each plan's slope mean? What does each starting value mean?**

**ANSWER / TEACHING NOTE**

Hours 0,1,2,3: Plan A costs 6,10,14,18; Plan B costs 10,13,16,19.

**2 | Which is cheaper at 1 hour? 3 hours? Explain using numbers and words.**

**ANSWER / TEACHING NOTE**

Slope A=$4/hr, start $6; slope B=$3/hr, start $10. Plan A is cheaper at 1 hour and 3 hours.

**Partner share:** Compare reasoning, not just answers. What would convince someone else your conclusion is correct?

# Writing & Interpreting Linear Models

## REVIEW & PREVIEW

**1 ★** A membership costs $5 per month plus $2 for each movie rented. Write a model for total monthly cost C for n movies. Interpret m and b.

**ANSWER / TEACHING NOTE**

C = 2n + 5. Rate $2/movie and starting fee $5/month.

**2** The table has points (0,15), (1,13), (2,11), (3,9). Write y = mx + b and interpret its trend.

**ANSWER / TEACHING NOTE**

m = -2, b = 15; y = -2x + 15.

**3 ★** Write the equation of a line through (0,4) and (2,10). Show work for m and b.

**ANSWER / TEACHING NOTE**

m = (10 - 4)/2 = 3; b = 4; y = 3x + 4.

# Writing & Interpreting Linear Models

## REVIEW & PREVIEW

**4** Choose which rule matches outputs 4, 7, 10 at inputs 0, 1, 2: (a) y=3x+4 (b) y=4x+3 (c) y=7x+4. Explain.

**ANSWER / TEACHING NOTE**

Choice (a): y = 3x + 4.

**5** Position is modeled by p = 20 - 5t (meters, seconds). Interpret 20 and -5; predict position at t = 3.

**ANSWER / TEACHING NOTE**

Initial position 20 m; velocity -5 m/s; p(3) = 5 m.

**6** Spiral + model check: Is {(0,7),(1,9),(2,11),(2,11)} a function? What linear rule fits these pairs?

**ANSWER / TEACHING NOTE**

Yes: the repeated pair is identical. Rule y = 2x + 7.

**Evidence check:** Star a question you can explain confidently. Circle one you want to revisit on Monday, November 9.